KOMTAS

DATA WAREHOUSE

ASSESSMENT

METHOD AND REFERENCE MODEL

Eleven dimensions, a five-level maturity ladder, an evidence model, and a target state decision framework

1

Positioning, Reference Model and Method

Where the engagement sits in the Data and Agentic AI value proposition, what is assessed, how each dimension is placed, what counts as evidence, the roles and capabilities required, sizing and schedule, the measurements taken, worked scenarios and anti-patterns.

p. 2 – 15

2

Scoping, Reporting and Target State

Scoping questions, how the result is reported, the keep / refactor / replace decision framework, decision log, glossary, evidence source mapping, delivery roadmap and the limits of the assessment.

p. 16 – 25

3

Client Findings

Current-state summary, target state and the delivery plan for one organisation. Filled in with the findings of the engagement.

p. 26 – 29

Where This Assessment Sits — AI Value Proposition

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Four layers. Agents are built in Agent 360 on top of everything below them; a data warehouse assessment is work in the bottom layer, and the condition the three above it depend on.

Action 360

Intelligent Work Experience

Teams | Slack | WhatsApp | Web / Portal | Claude Desktop | API, Headless Interfaces

Agent 360

Governed Digital Workforce

Agents solve a problem in one business line

BY INDUSTRY: Manufacturing | Retail | Banking | Energy | Aviation | Other industries

BY FUNCTION: Customer Services | Sales | Finance | Procurement | Operations | Other agents

Context 360

Enterprise Context

Semantic Layer | Ontology | Knowledge Graph | Master Data | Corporate Memory

Data / Analytic 360

Trusted Data Foundation

THIS ENGAGEMENT · 11 DIMENSIONS

Datawarehouse / Lakehouse | Integration | Data Virtualization | Data Governance / MDM | Lineage | Access Control

Each layer consumes the one below it. The assessment covers the whole bottom layer across eleven dimensions and places each one on evidence.

WHAT THE FOUNDATION DETERMINES ABOVE IT

Context 360

A semantic layer, an ontology and a knowledge graph inherit the warehouse's definitions, lineage and quality. Built on an unassessed foundation, they encode its errors as meaning.

Agent 360

An agent solving a problem in one business line is only as reliable as the data it acts on. Agents turn a data quality problem into an action problem.

Action 360

Delivery through Teams, Slack, a portal or an API multiplies exposure to whatever the foundation gets wrong, across every channel at once.

Data Warehouse Assessment — Reference Model

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What is examined, where it is examined, and what kind of evidence it produces

Platform- and vendor-independent. The model defines what is assessed and what evidence closes it; it does not assume a target product.

1. Sources and Ingestion

- **Source inventory** — systems, owners, volumes
- **Ingestion pattern** — batch, CDC, streaming
- **Load windows** — duration vs. available window
- **Pipeline reliability** — success rate, MTTR
- **Onboarding lead time** — days to add a source

2. Storage and Layering

- **Layer definition** — raw, cleansed, curated
- **Storage footprint** — growth, compression, copies
- **Retention and archive** — policy vs. actual
- **Partitioning and indexing**
- **Duplication across layers** — how many copies exist

3. Modelling and Semantics

- **Modelling approach** — dimensional, vault, flat
- **Historisation** — slowly changing dimensions
- **Semantic layer** — where metrics are defined
- **Naming and standards**
- **Model coverage** — critical entities modelled

4. Serving and Consumption

- **Workload classes** — BI, ad hoc, extract, ML
- **Query latency** — p95 by workload class
- **Concurrency** — peak vs. capacity
- **Certified vs. shadow reports**
- **Self-service enablement**

5. Quality and Reconciliation

- **Reconciliation to source** — break rate vs. source system
- **Quality rules** — coverage and enforcement
- **Reload and exceptions** — rejected record path, repeatability
- **Freshness vs. SLA** — per critical dataset
- **Schema change breakage** — frequency and blast radius
- **Trust signals** — what consumers actually believe

6. Operations

- **Monitoring and alerting**
- **Incident history** — volume and causes
- **Change management** — release and rollback
- **Disaster recovery** — last successful test

7. Governance

- **Ownership** — named owner per dataset
- **Lineage** — end to end coverage
- **Catalogue and metadata**
- **Access and privacy** — classification, reviews

Cross-Cutting Dimensions — assessed across every column

Cost and Efficiency

- Cost per TB stored and scanned
- Idle vs. active compute ratio
- Chargeback and visibility

Security and Privacy

- Classification of sensitive columns
- Access review coverage
- Masking and row-level rules

Performance and Scale

- Headroom against growth
- Elasticity and workload isolation
- Degradation under concurrency

People and Operating Model

- Roles and separation of duties
- Key person dependency
- Skill coverage per platform

AI and ML Readiness

- Feature and training data access
- Unstructured data position
- Reproducibility of a dataset

Evidence and Measurement

- Metric definitions agreed in advance
- Observation window stated
- Three-state evidence scale

NOTES

- The columns are a map of what is examined, not a sequence of work.
- Every item resolves to either a measurement or a stated evidence gap.
- Scope may be narrowed; narrowing is written into the report, never left implicit.

COLOUR LEGEND — ITEM TYPE

- Structure
- Governance
- Measurement
- Cost & Consumption

HOW TO USE THIS MODEL

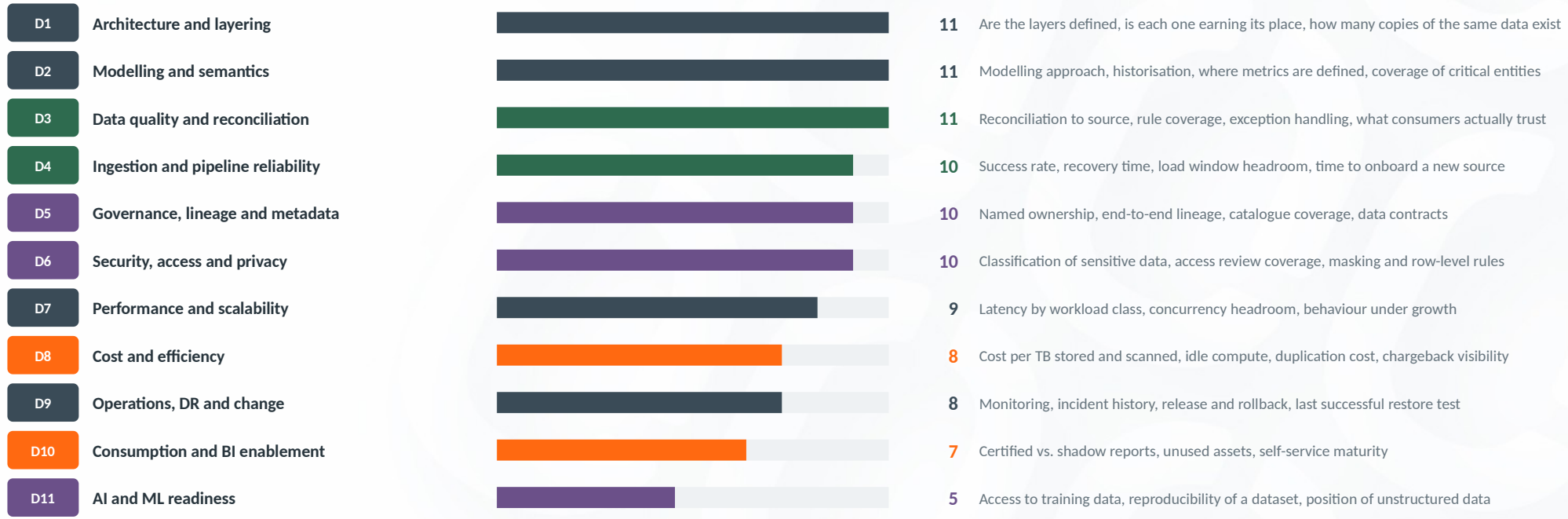
A catalogue of what a full assessment covers. The engagement scope selects from it; what is left out is declared, not dropped.

Assessment Dimensions and Weights

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Eleven dimensions, weights summing to 100. The weights are fixed before the engagement starts and do not change with what the assessment finds.

DIMENSION WEIGHTS



D1	Architecture and layering	11	Are the layers defined, is each one earning its place, how many copies of the same data exist
D2	Modelling and semantics	11	Modelling approach, historisation, where metrics are defined, coverage of critical entities
D3	Data quality and reconciliation	11	Reconciliation to source, rule coverage, exception handling, what consumers actually trust
D4	Ingestion and pipeline reliability	10	Success rate, recovery time, load window headroom, time to onboard a new source
D5	Governance, lineage and metadata	10	Named ownership, end-to-end lineage, catalogue coverage, data contracts
D6	Security, access and privacy	10	Classification of sensitive data, access review coverage, masking and row-level rules
D7	Performance and scalability	9	Latency by workload class, concurrency headroom, behaviour under growth
D8	Cost and efficiency	8	Cost per TB stored and scanned, idle compute, duplication cost, chargeback visibility
D9	Operations, DR and change	8	Monitoring, incident history, release and rollback, last successful restore test
D10	Consumption and BI enablement	7	Certified vs. shadow reports, unused assets, self-service maturity
D11	AI and ML readiness	5	Access to training data, reproducibility of a dataset, position of unstructured data

Why weights at all

Without them, the loudest finding sets the agenda. A weight states before the engagement which dimension carries more consequence for this organisation.

When a weight may change

Only at scoping, with the client, and written into the report. A weight is never adjusted after the findings are known.

What a weight is not

It is not a budget allocation and not a measure of effort. A dimension may score well and still consume most of the remediation cost.

Maturity Ladder — How Each Dimension Is Placed

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Each dimension is placed on the same five-level ladder. A level is claimed only where evidence supports it; without evidence the level below is recorded.

1 · Ad hoc

It works because particular people make it work.

- No written definition of layers or ownership
- Loads are fixed by hand when they fail
- Numbers are reconciled in spreadsheets
- Nobody can say what a dataset costs

2 · Repeatable

The same thing is done the same way, but only by habit.

- Conventions exist in people's heads
- Pipelines rerun but failures are found by users
- Some quality checks, no coverage figure
- Documentation lags behind the platform

3 · Defined

Written down, agreed, and the same across teams.

- Layers, ownership and naming are documented
- Load SLAs stated per critical dataset
- Quality rules versioned with the pipeline
- Lineage exists for the critical path

4 · Managed

Measured against targets and acted on when it drifts.

- Freshness, latency and cost are monitored against targets
- Incidents have causes and closure times
- Access reviews run on a cycle
- Change goes through release and rollback

5 · Optimised

The platform improves itself on evidence, not on opinion.

- Cost and performance tuned against measured workloads
- Unused assets retired on a cycle
- New sources onboarded to a known lead time
- Model and semantics evolve with a contract

A dimension is placed at the highest level for which every statement below it also holds. A platform that monitors cost but has no written ownership is level 2, not level 4 — the ladder is not a menu.

Level 5 is not the target for every dimension. The target level is set per dimension against what the organisation needs; pushing a dimension above its needed level is spend without a return.

Evidence Model — What Closes a Question

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An assessment is worth what its evidence is worth. The evidence type is recorded next to every finding, so the reader can see which conclusions rest on measurement and which rest on a claim.

1. MEASUREMENT

A number produced from the platform itself over a stated window.

- Query log extract
- Pipeline run history
- Storage and compute billing records
- Catalogue and lineage export

2. ARTEFACT

A document or configuration that exists independently of what anyone says.

- Data model and DDL
- Pipeline and job definitions
- Access control configuration
- Runbooks and DR test records

3. OBSERVATION

Something seen working, or failing, in a walkthrough.

- Load executed end to end
- A report traced back to source
- A failed job recovered
- An access request fulfilled

4. STATEMENT

What a person says. Recorded as a claim, never as a finding on its own.

- Interview and workshop notes
- Self-assessment questionnaires
- Vendor and internal roadmaps
- Historical context not in any system

A statement alone never raises a maturity level. “We reconcile every night” is recorded as a claim until a reconciliation run or its output is seen; where it cannot be seen, the report says so rather than resolving it in either direction.

THREE-STATE SCALE — APPLIED PER QUESTION, NOT PER DIMENSION

Evidence provided

A measurement, artefact or observation supports the stated level.

Partial evidence

Part of the evidence exists; the remainder was out of scope or unavailable.

Evidence not produced

The evidence could not be produced in the engagement window. It does not mean the control is absent.

RULES APPLIED TO THE EVIDENCE ITSELF

An absent measurement is a finding

If the platform cannot report its own query latency or load success rate, that is itself a maturity signal and is recorded as one.

The observation window is stated

A two-week window during a quiet period does not describe peak behaviour. The window is written next to every measurement.

Scope reductions stay visible

Anything excluded for access, time or confidentiality reasons is listed. A narrowed scope lowers confidence, it does not raise the score.

Role Boundaries in the Assessment

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An assessment fails on access and on ownership of the conclusion, not on method. Both are settled before the first interview.

ROLE	SIDE	DOES	DOES NOT
Assessor	External	Collects evidence · measures · places each dimension on the ladder · writes the report	Does not change anything on the platform · does not accept a claim as a finding
Platform / Data Engineering	Client	Provides access, logs and configuration · explains how the platform actually runs	Does not pre-filter the evidence · does not decide which findings reach the report
Data Owners	Client	Confirm which datasets are critical and what their freshness and quality targets are	Do not set the weights after the findings are known
Business / Consumers	Client	State what they rely on, what they do not trust, and what they rebuild by hand	Are not asked to grade the platform; their input is a measurement of trust, not a score
Security and Compliance	Client	Confirm classification, access rules and what may leave the environment	Do not narrow scope silently; every exclusion is written into the report
Sponsor	Client	Sets the scope, the target levels per dimension and the decision the assessment must support	Does not change the target level after seeing the current level
Steering / Validation	Joint	Reviews findings for factual accuracy before the report is issued	Corrects facts, not conclusions · disagreement is recorded, not removed

The assessor owns the findings; the client owns the decision. Where the two are mixed — the client editing findings, or the assessor choosing the target state alone — the report stops being usable as a basis for investment.

Assessment Process

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Five phases, roughly six to nine weeks for a mid-sized estate. The phases are sequential because each one constrains what the next is allowed to claim.

P0 · Scoping

1 week

- Dimensions in scope and target level per dimension
- Critical datasets and workloads named by the client
- Access requests raised and granted
- Metric definitions and observation window agreed in writing

RULE — Nothing is measured until the metric definitions are agreed; otherwise the numbers are argued, not used

P1 · Evidence collection

2 – 3 weeks

- Query logs, pipeline history, billing and catalogue exports
- Data model, DDL, job definitions and access configuration
- Walkthroughs: a load, a report traced to source, a recovery
- Interviews with owners, engineers and consumers

RULE — Each item lands as measurement, artefact, observation or statement — the type is recorded with it

P2 · Measurement and placement

1 – 2 weeks

- Metrics computed over the agreed window
- Each dimension placed on the ladder against the evidence
- Gap between current and target level stated per dimension
- Weighted score computed and the composition shown

RULE — A level without evidence is not claimed; the level below is recorded with the gap stated

P3 · Target state and roadmap

1 – 2 weeks

- Target architecture decision against the criteria
- Findings grouped into remediation themes with sequence
- Effort, dependency and risk per theme
- What is deliberately not being fixed, and why

RULE — Every recommendation traces to a finding; a recommendation with no finding behind it is removed

P4 · Validation and issue

1 week

- Factual review with the platform team
- Disagreements recorded alongside the finding, not resolved away
- Report issued with scope, window and evidence gaps declared
- Baseline stored for the next assessment

RULE — A single assessment is a snapshot; the value appears on the second one

Access is the critical path, not analysis. Where read access to logs, billing and catalogue cannot be granted, the schedule is set from what access is actually available — not from the ideal plan.

An assessment that ends at a score has not finished. The deliverable is a decision: what to keep, what to refactor, what to replace, and in which order.

Capabilities Required — By Role

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The roles the engagement needs, on both sides. A missing role is not absorbed by the others; it becomes a dimension that is assessed on statements rather than evidence.

ROLE	SIDE	WHAT IT CONTRIBUTES	PHASES	INVOLVEMENT
Assessment Lead	Komtaş	Owns the method, the scope and the report. Places dimensions, writes findings, defends them in the readout.	**P0–P4**	Throughout
Data Architect	Komtaş	Assesses architecture, layering, modelling and the target state options. Produces the keep / refactor / replace recommendation.	**P1–P3**	Throughout
Platform / DWH Engineer	Komtaş	Extracts and computes the measurements: query logs, run history, billing, schema. Knows the specific platform in use.	**P1–P2**	Heaviest in P1
Data Governance Consultant	Komtaş	Assesses ownership, lineage, catalogue, contracts and classification. Carries the Context 360 conversation.	**P1–P3**	Part-time
BI / Analytics Consultant	Komtaş	Assesses consumption: certified vs. shadow, usage, semantic definitions, where numbers are rebuilt by hand.	**P1–P2**	Part-time
Security and Privacy Specialist	Komtaş	Assesses classification, access reviews, masking and residency constraints.	**P1**	Short, focused
FinOps / Cost Analyst	Komtaş	Assesses attributable cost, idle capacity and duplication. Often the finding that funds the programme.	**P1–P2**	Short, focused
Client Sponsor	Client	Sets the decision the assessment supports, the scope and the target level per dimension. Clears access.	**P0, P3–P4**	Decision points
Client Platform Lead	Client	Grants access, explains how the platform actually runs, validates facts before issue.	**P0–P1, P4**	Throughout
Client Data Owners	Client	Name the critical datasets, their freshness and quality targets, and confirm what they are accountable for.	**P0–P1**	Workshops only

Minimum viable team

Assessment Lead, Data Architect and Platform Engineer. Below this, the measurement set is too thin to place eleven dimensions on evidence.

Where the gap usually is

Governance and FinOps. Both are treated as optional and both produce the findings that change the sponsor's decision.

Client-side is not optional

Without a platform lead who can grant access and explain how the estate really runs, the engagement becomes interview-based and says so in its confidence rating.

Sizing the Engagement — the Metrics That Set the Schedule

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Eleven drivers, collected in P0 before any commitment on dates. Duration follows the estate and the access, not the ambition of the scope.

SIZING DRIVER	SMALL	MEDIUM	LARGE
Source systems feeding the warehouse	S up to 15	M 16 – 50	L more than 50
Critical datasets in scope	S up to 30	M 31 – 100	L more than 100
Subject areas	S up to 3	M 4 – 8	L more than 8
Distinct warehouse / lake platforms	S one	M two	L three or more
Reports and dashboards in the estate	S up to 300	M 301 – 1,500	L more than 1,500
Environments to be examined	S production only	M production and one other	L three or more
Legal entities or regions in scope	S one	M two or three	L more than three
Access lead time for logs and billing	S granted within a week	M two to three weeks	L longer, or partial only
Regulatory exposure	S none specific	M sector-regulated	L multi-jurisdiction
Log and usage history retention	S 90 days or more	M 30 – 89 days	L under 30 days
Prior assessment baseline	S exists and is usable	M partial	L none

BANDING RULE — the engagement takes the highest band that at least three drivers fall into. A single large driver does not move the band; three do, because they compound.

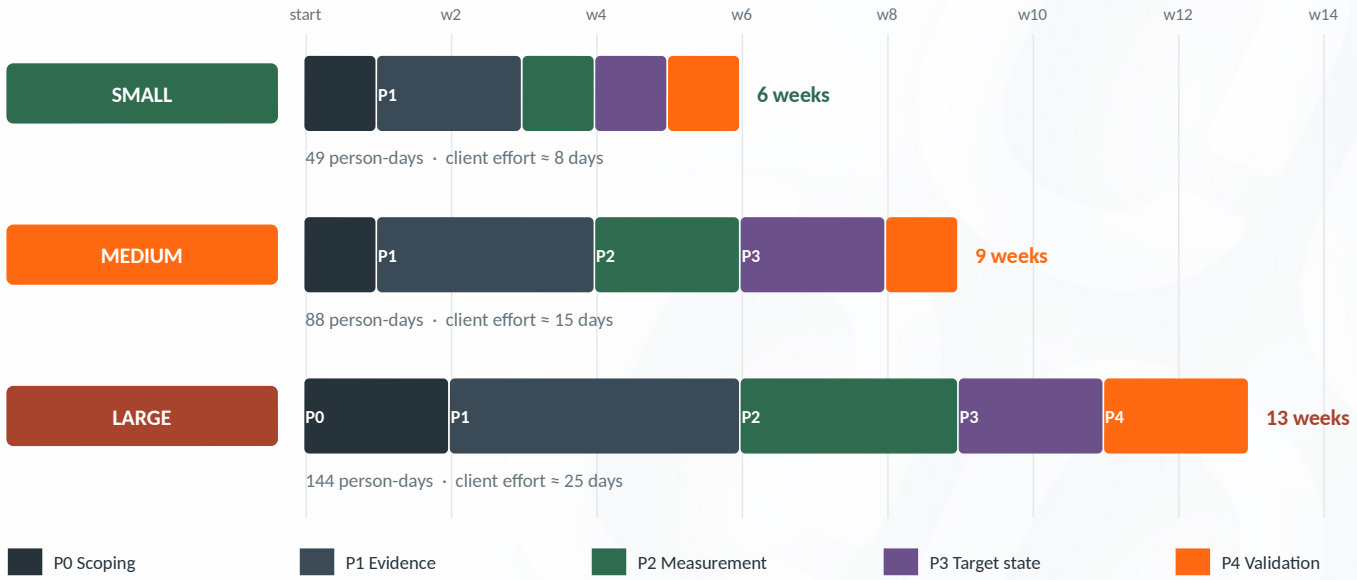
Access lead time and log retention are the two drivers that cannot be compensated for by adding people. Where either lands in the large band, the date moves regardless of team size.

Estimated Schedule and Effort

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Indicative durations by band, from the sizing drivers. Committed dates are set in P0 once access lead times are known, not before.

TIMELINE BY BAND



Phases overlap by at most a few days. P2 cannot begin on a dataset whose evidence is still being collected, and P3 cannot begin before placement is complete.

These are estimates for the assessment only. Stage 2 foundations and Stage 3 platform work are sized separately, after the findings, because their scope is what the assessment is there to establish.

Effort figures assume read-only exports over an agreed window rather than standing access, and a client platform lead available throughout P1.

EFFORT BY ROLE — PERSON-DAYS

ROLE	S	M	L
Assessment Lead	15	25	40
Data Architect	10	18	30
Platform / DWH Engineer	10	18	28
Data Governance Consultant	5	10	16
BI / Analytics Consultant	4	8	14
Security and Privacy Specialist	2	4	7
FinOps / Cost Analyst	3	5	9
TOTAL	49	88	144

WHAT MOVES THE DATE

Access approvals	The dominant driver. Each evidence source is a separate approval; raised as one list in P0.
Short log retention	If history is under 30 days, an instrumentation window is added before P1 can produce a usable measurement.
Multiple platforms	Each additional platform repeats extraction and measurement; it does not repeat the method.
Unavailable data owners	Critical datasets cannot be named by the platform team on the owners' behalf; workshops are rescheduled, not skipped.
Scope added after P0	New subject areas re-open the sizing. The band is recalculated and the date is reset with it.

Measurements the Assessment Takes

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The quantitative spine. Each metric is defined and agreed at scoping, computed over a stated window, and reported with the window next to it.

MEASUREMENT	HOW IT IS OBTAINED	WHAT IT TELLS	TYPICAL RED FLAG
PIPELINE AND FRESHNESS			
Load success rate and MTTR	Scheduler / orchestrator run history	Whether the platform recovers itself or people recover it	Recovery measured in people, not minutes
Freshness lag vs. stated SLA	Load completion time per critical dataset	Whether the promise made to consumers is kept	An SLA exists but was never measured
Source onboarding lead time	Date raised to date in production, last 10 sources	Whether adding data is routine or a project each time	Measured in quarters
PERFORMANCE AND CAPACITY			
Query latency p95 by workload class	Query log over the agreed window	Whether the platform serves the work it is actually asked to do	One class starved by another
Concurrency at peak vs. capacity	Query log, concurrency slots or queue depth	Where the ceiling is and how close daily use gets to it	Queueing is normal and nobody reports it
Growth headroom	Load duration vs. window; storage growth vs. capacity	How long the current design survives at current growth	Batch window above 80% used, no plan for growth
QUALITY AND TRUST			
Reconciliation break rate	Comparison of warehouse totals to source totals	Whether the warehouse agrees with the systems it reports on	Breaks are known and tolerated
Quality rule coverage	Rules defined vs. critical datasets	How much of the critical path is actually checked	Rules concentrated where failures are harmless
Schema change breakage	Change history vs. downstream incidents	Whether change is absorbed or propagated as outage	No record of what broke last time
COST AND FOOTPRINT			
Cost per TB stored and scanned	Billing records against workload tags	Where the money goes and whether anyone can see it	Cost not attributable to any team or workload
Idle vs. active compute	Utilisation over the window	How much is paid for capacity nobody used	Always-on clusters serving a daily batch
Copies of the same data	Layer inventory and lineage	How much of the footprint is duplication rather than value	The same table in four places, three unowned
GOVERNANCE AND ACCESS			
Ownership and contract coverage	Catalogue export against critical dataset list	Whether anyone is accountable for a dataset	Critical datasets with no named owner
Lineage completeness	Lineage export vs. critical reports	Whether impact of a change can be assessed before making it	Lineage stops at the warehouse boundary
Access review coverage	Access control configuration and review records	Whether permissions reflect current roles	Last review predates a reorganisation
CONSUMPTION			
Certified vs. shadow reports	BI platform inventory and usage logs	How much of consumption runs outside the governed path	Extracts rebuilt by hand in spreadsheets
Unused assets	Usage logs over 90 days	How much of the estate is maintained for nobody	A third of reports unopened, none retired

Where a measurement cannot be taken — no query log retention, no workload tagging, no usage history — that absence is itself recorded as a finding. A platform that cannot describe its own behaviour cannot be managed against a target.

Scenario — Assessing Pipeline Reliability and Freshness

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One dimension, start to finish. The same shape applies to the other ten: scope what matters, collect evidence of a stated type, measure, place, conclude.

STEP	PHASE	WORK PERFORMED	OUTPUT
SCOPE · what matters and to whom			
1	Scope	The client names the datasets whose freshness actually matters to a decision	Critical dataset list
2	Scope	The stated SLA for each one is written down, with who it was promised to	SLA statements
EVIDENCE · collection			
3	Evidence	Orchestrator run history is exported over the agreed window	Run history extract
4	Evidence	Load completion timestamps are compared to the SLA, per dataset per day	Freshness lag series
5	Evidence	Failed runs are grouped by cause and by how they were recovered	Failure taxonomy
6	Evidence	A load is observed end to end, including a deliberate rerun	Observation record
MEASUREMENT			
7	Measure	Success rate, mean time to recovery and lag distribution are computed	Three measurements
8	Measure	Load duration is compared to the available window to find remaining headroom	Headroom figure
9	Measure	Consumers are asked what they believe the freshness is	Perceived vs. actual gap
PLACEMENT			
10	Place	The dimension is placed on the ladder against the evidence, not the intent	Level with gap stated
11	Place	Where a measurement could not be taken, the gap is recorded instead of estimated	Declared evidence gap
CONCLUSION			
12	Conclude	Findings are tied to consequence: which decision is made on stale data	Business impact
13	Conclude	Remediation is sequenced by consequence, not by ease	Ordered remediation

Step 9 is not a courtesy. The gap between what consumers believe the freshness is and what it actually is predicts where numbers get rebuilt by hand, and it is usually the finding that moves the sponsor.

Scenario — From Finding to Recommendation

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Every recommendation is traceable backwards to a finding, and every finding forwards to a consequence. A recommendation with no finding behind it does not reach the report.

STEP	STAGE	WORK PERFORMED	OUTPUT
FINDING · what was seen			
1	Finding	Twelve critical datasets have no named owner	Catalogue export vs. critical list
2	Evidence	Artefact — the catalogue itself, not an interview answer	Evidence type recorded
IMPACT · why it matters here			
3	Impact	A schema change last quarter broke four reports; nobody was notified in advance	Incident record
4	Impact	Consequence stated in the client's own terms, not in generic risk language	Business consequence
PLACEMENT			
5	Placement	Governance dimension placed at level 2; target level for this client is 4	Level and gap
RECOMMENDATION			
6	Recommendation	Assign owners for the critical list before any platform change	Action with a sequence
7	Recommendation	Make ownership a precondition in the catalogue, not a convention	Structural fix
8	Dependency	Ownership must precede the lineage work, which precedes contract enforcement	Sequence constraint
TARGET STATE AND SCOPE			
9	Target state	Does this finding change the target architecture decision, or only the path to it	Decision linkage
10	Not doing	Full catalogue coverage is out of the first phase; stated and justified	Declared exclusion

Step 10 matters as much as the recommendations. A report that lists everything worth doing and nothing worth postponing gives the sponsor no way to start.

What Not to Do in an Assessment

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Common ways an assessment loses its value; why each looks reasonable at the time, what it breaks, and what replaces it.

WHAT NOT TO DO	WHY IT LOOKS REASONABLE	WHAT BREAKS	INSTEAD
Grading the platform from interviews	People are available, logs are not; interviews feel like progress	The assessment measures confidence rather than the platform; the loudest team scores best	**Every level is backed by a measurement, artefact or observation; statements are recorded as claims**
Agreeing metric definitions after seeing the numbers	The definition debate feels academic until there is a number to argue about	Each finding is re-litigated on definition; the report loses authority at exactly the point it is needed	**Metric definitions and the observation window are agreed in writing at scoping**
Measuring a quiet window	A short, calm window is easier to get access to and produces clean data	Peak behaviour — where the platform actually fails — is never observed	**The window covers a period the client agrees is representative, and the window is printed with the number**
Benchmarking against an industry average	An external number looks objective and is easy to present	The comparison hides whether the platform meets this organisation's own needs	**Each dimension is compared to a target level the client set for themselves at scoping**
Letting the score become the conclusion	A single number travels well in a steering committee	A low-weight dimension with a severe finding disappears behind a respectable total	**The score is reported with its composition and never without the findings that produced it**
Recommending a platform before the findings are placed	The target product is often decided before the assessment starts	The assessment becomes a justification exercise and stops being usable as evidence	**The target state decision is made in P3, against stated criteria, after placement**
Removing findings the client disputes	Keeping the relationship comfortable through the readout	The report becomes a negotiated document; the next assessment has no honest baseline	**Facts are corrected; disagreement is recorded next to the finding and both are issued**
Issuing a report with no second assessment planned	The engagement has a defined end and the invoice is sent	There is no baseline, so no improvement can ever be demonstrated	**The finding set is stored as a dated baseline and a re-assessment point is agreed**

Each of these produces a report that reads well. The cost appears when someone tries to use it — to fund remediation, to defend a platform decision, or to show that anything improved.

Scoping Questions — Decision, Scope and Access

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Asked before anything is measured. The answers set the engagement boundary and determine which measurements are possible.

#	QUESTION	WHAT IT DETERMINES	RESULT BY ANSWER
1	**What decision is this assessment meant to support**	**The whole shape of the engagement**	No decision named: the report will be read and filed; scope the engagement around one
2	**Which datasets and reports are critical, and who says so**	**Where measurement effort goes**	If nobody can name them, that is the first finding and the first workshop
3	**What is the target maturity level per dimension**	**What counts as a gap**	Without a target, every finding looks equally urgent and nothing gets sequenced
4	**Can read access to query logs, run history and billing be granted**	**Whether measurement is possible at all**	If not, the assessment is interview-based and its confidence is reported as such
5	**How long is query and usage history retained**	**The observation window**	Short retention caps the window; the cap is printed next to every measurement
6	**Are workloads tagged or attributable to teams**	**Whether cost can be assessed**	Untagged compute means cost findings stay at platform level, not workload level
7	**Which platform and deployment model is the warehouse on**	**Which measurements are available and how**	Determines the extraction method; it does not change the dimensions or weights
8	**How many source systems feed it and by what pattern**	**Ingestion scope**	A long tail of one-off loads is a different finding from a few heavy pipelines
9	**Is there a semantic layer, or are metrics defined in reports**	**Modelling and consumption scope**	Metrics defined in reports predicts disagreement between numbers; measure the duplication
10	**Has an assessment been done before, and is its output available**	**Whether a baseline exists**	A prior finding set turns this engagement from a snapshot into a trend
11	**What is out of bounds — systems, data, environments**	**Declared scope reduction**	Every exclusion is listed in the report; an undeclared exclusion invalidates the comparison

These questions are asked at scoping, before any measurement. Their purpose is to size the engagement and to establish what the client will accept as evidence.

Scoping Questions — Governance, Trust and Target

KOMTAS

Asked at scoping and revisited during evidence collection, because the honest answers usually arrive in the second conversation.

#	QUESTION	WHAT IT DETERMINES	RESULT BY ANSWER
12	**Who owns the critical datasets today, by name**	**Governance scope**	Role-level answers without names mean ownership is nominal; verify against the catalogue
13	**Does lineage exist, and where does it stop**	**Change impact capability**	Lineage that stops at the warehouse edge cannot support an impact assessment
14	**How are schema changes released and communicated**	**Change management maturity**	No release path means downstream breakage is discovered by consumers
15	**What is reconciled to source, how often, and by whom**	**Trust in the numbers**	Manual spreadsheet reconciliation is both a finding and a measurement of distrust
16	**Which reports are certified, and how many exist in total**	**Consumption governance**	A large uncertified estate is where the real consumption happens; measure usage, not inventory
17	**Where do people rebuild numbers outside the platform**	**The honest trust signal**	Shadow extracts locate exactly which datasets are not believed
18	**Is sensitive data classified, and are access reviews run**	**Security and privacy scope**	No classification means masking and row-level rules cannot be assessed as effective
19	**When was the last successful restore or failover test**	**Operational resilience**	A DR design with no test record is a claim, not a capability
20	**What does the platform cost, and who sees that number**	**Cost dimension feasibility**	If only finance sees the bill, cost findings have no owner on the technical side
21	**Is ML or AI work already consuming warehouse data, and how**	**AI readiness scope**	Extracts copied to notebooks indicate the serving layer does not meet that workload
22	**What has already been decided about the target platform**	**Whether the decision is genuinely open**	If the target is fixed, the engagement is a readiness assessment; it is scoped and named as one

An unanswered question is an assumption. Assumptions are written into the engagement letter and into the report, so that a result that changes between two assessments can be attributed to the platform rather than to a shifted assumption.

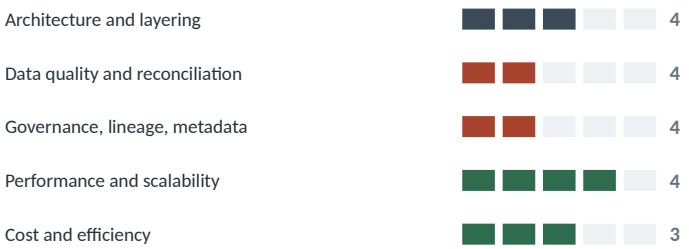
How the Result Is Reported

KOMTAS

Three outputs, read together. The dimension view drives remediation, the gap view drives sequencing, and the composite exists only to show movement between assessments.

1 · DIMENSION VIEW

Each of the eleven dimensions with its current level, target level and the findings behind it.



illustrative — bars show current level, figure shows target

2 · GAP AND SEQUENCE

Dimensions ordered by weighted gap, with dependencies, to produce a remediation sequence.

- Weighted gap = (target − current) × dimension weight
- Dependencies override raw gap order: ownership before lineage, lineage before contracts
- Each theme carries effort, dependency and risk
- What is deliberately not being fixed is listed with it

3 · COMPOSITE SCORE

Weighted sum of the eleven dimension levels, on a 0–100 scale. Reported with its composition, never alone.

Its only legitimate use is comparison with this organisation's own previous assessment.

- Not a benchmark against other organisations
- Not a pass or fail threshold
- Not a substitute for the findings

A composite score is the part of the report that travels furthest and carries the least information. It is reported because it is the only thing that makes two assessments comparable at a glance — and for no other reason.

WHAT IS HANDED OVER

Assessment report

Findings by dimension with evidence type, current and target level, and the gap. The working document.

Executive summary

The decision the assessment was commissioned to support, the three findings that bear on it, and the recommendation.

Remediation plan

Themes in sequence with dependency, effort band and risk, including what is not being addressed and why.

Target state recommendation

Keep, refactor, replace or hybrid — against the stated criteria, with the rejected options and their reasons.

Evidence pack

The measurements, exports and observation records the findings rest on, retained as the dated baseline.

Target State Decision — Keep, Refactor or Replace

KOMTAS

Decided in P3, after placement, against criteria agreed at scoping. No single criterion decides; the pattern across them does.

CRITERION	POINTS TO KEEP AND OPTIMISE	POINTS TO REFACTOR IN PLACE	POINTS TO REPLACE / LAKEHOUSE
Workload the platform must serve	BI and reporting on modelled data	Mixed BI, exploration and ad hoc SQL	BI plus ML, streaming and semi-structured
Data types in scope	Structured, relational	Mostly structured, some semi-structured	Structured, semi-structured and unstructured
Scale and growth trajectory	Predictable, within current capacity	Growing but bounded	High or unpredictable growth
Concurrency profile	Known peaks, stable user base	Variable, bursty	Heavily variable with isolated workload classes
Cost model requirement	Predictable fixed cost preferred	Mixed, some elasticity needed	Consumption-based, elastic
Engineering capability	SQL and ETL tooling	SQL plus some engineering	Engineering-led, code-first
Governance maturity today	Level 3 or above	Level 2 to 3	Level 3 or above required before the move
Existing investment	Significant, recent, serving its purpose	Significant but constraining	Reaching end of life or end of contract
Regulatory and residency constraints	Satisfied today	Satisfiable with work	Requires explicit verification on the target
Time pressure	No external deadline	Deadline in 12 months or more	Deadline inside 12 months

The default recommendation is to fix the operating model before replacing the platform. Ownership, lineage and quality gaps follow the data onto any new platform and arrive there with migration risk added. Replacement is recommended where the platform itself is the binding constraint.

KEEP AND OPTIMISE

The platform is not the constraint; the operating model is.

REFACTOR IN PLACE

The platform is sound, the design on it is not.

REPLACE OR MOVE TO LAKEHOUSE

The platform cannot serve the workload the organisation now has.

Decision Log — Method and Evidence

KOMTAS

For each decision: the rationale, the rejected alternative, and under which condition it is revisited.

#	TOPIC	DECISION TAKEN	RATIONALE AND ALTERNATIVE	WHAT WOULD CHANGE IT
1	**Basis of a level**	**Evidence only — measurement, artefact or observation**	A level granted on a statement cannot be re-tested at the next assessment. Rejected: self-assessment questionnaires as a basis for placement.	Does not change — it is what makes two assessments comparable
2	**Weights**	**Fixed at scoping, with the client, never after findings**	Adjusting weights once the results are known turns the score into an output anyone can shape. Rejected: weighting by where the findings turned out to be.	A material change in the organisation's priorities, agreed before the next assessment
3	**Target levels**	**Set per dimension by the client, not by the assessor**	Level 5 everywhere is spend without a return; the needed level differs by dimension. Rejected: a single target level for the whole platform.	A change in regulatory exposure or in what the business runs on the platform
4	**Benchmarking**	**Against the client's own target levels, not an industry average**	An external average says nothing about whether this platform meets this organisation's needs. Rejected: peer comparison as the headline of the report.	A specific regulatory or contractual benchmark that the client must meet
5	**Score reporting**	**Always with its composition and the findings behind it**	A composite number travels further than the findings and starts being used alone. Rejected: a single headline score on the summary page.	Does not change — the composition is part of the number
6	**Disputed findings**	**Facts corrected, disagreement recorded, both issued**	A negotiated report leaves no honest baseline for the next assessment. Rejected: removing findings the platform team disputes.	Does not change — a factual error is corrected, a disagreement is published
7	**Measurement gaps**	**Recorded as findings, not estimated**	An estimate presented as a measurement cannot be re-tested and quietly becomes fact. Rejected: filling gaps with reasonable assumptions.	Access is granted later and the measurement is taken; the report is versioned

Decision Log — Target State and Reporting

KOMTAS

For each decision: the rationale, the rejected alternative, and under which condition it is revisited.

#	TOPIC	DECISION TAKEN	RATIONALE AND ALTERNATIVE	WHAT WOULD CHANGE IT
8	**Target architecture**	**Decided in P3 against stated criteria, after placement**	Choosing the target first turns the assessment into a justification exercise. Rejected: starting from a product decision already taken.	The client states the decision is already made; the engagement is then scoped as readiness, not assessment
9	**Default recommendation**	**Fix the operating model before replacing the platform**	Ownership, lineage and quality gaps follow the data to any new platform. Rejected: migration as the default answer to a low score.	Where the platform itself is the binding constraint — capacity, cost model or workload class it cannot serve
10	**Remediation sequence**	**Ordered by consequence, with dependencies explicit**	A list ordered by ease produces visible motion and no change in outcome. Rejected: quick wins first as a general principle.	A hard external deadline that forces a specific item forward
11	**What is not being fixed**	**Stated explicitly with the reason**	A report with no exclusions gives the sponsor no way to start. Rejected: listing everything worth doing without priority.	Does not change — the exclusion list is part of the deliverable
12	**Cost findings**	**Reported as attributable cost, not as a saving promise**	A saving figure becomes a commitment the assessment cannot stand behind. Rejected: headline savings estimates in the summary.	A costed remediation plan is a separate engagement with its own assumptions
13	**Baseline**	**Finding set stored, dated, with the window and scope**	Improvement cannot be demonstrated against a report nobody can reproduce. Rejected: issuing a document with no retained finding set.	Does not change — retention period follows the client's policy
14	**Re-assessment**	**A re-assessment point is agreed at issue**	The second assessment is where the method starts producing a trend. Rejected: treating the engagement as complete at report issue.	The cadence follows the remediation plan, typically six to twelve months

Glossary

KOMTAS

Shared definitions of the terms used in the assessment and the report. Used to separate a debate about terms from a debate about the platform.

Assessment dimension
One of the eleven areas the platform is examined in, each carrying a fixed weight.

Maturity level
The position of a dimension on the five-level ladder, claimed only where evidence supports it.

Target level
The level this organisation needs in a given dimension, set by the client at scoping.

Evidence type
Whether a conclusion rests on a measurement, an artefact, an observation or a statement.

Observation window
The period a measurement covers. Printed next to the measurement, because it bounds what it means.

Critical dataset
A dataset the client names as carrying a decision or an obligation. Measurement effort concentrates here.

Freshness lag
The delay between a fact occurring in the source and being visible in the warehouse.

Load window
The time available for a batch to complete before consumers need the data.

MTTR
Mean time to recovery — how long a failed load stays failed, measured from history rather than from design.

Reconciliation break
A difference between warehouse totals and source totals that the platform does not explain.

Data contract
A written agreement between producer and consumer on schema, meaning, freshness and quality.

Lineage
The end-to-end path of a value from source through transformations to a report. The basis of impact analysis.

Semantic layer
Where metric definitions and access rules live independently of any single reporting tool.

Historisation
How a changing attribute is retained over time. The precondition for a correct point-in-time report.

Serving layer
The modelled layer consumption reads from, as opposed to the layers it is built from.

Workload class
A group of queries with shared characteristics — BI, ad hoc, extract, ML — measured separately.

Concurrency headroom
The distance between peak concurrent demand and the platform's capacity to serve it.

Shadow report
Consumption built outside the governed path. Locates the datasets that are not believed.

Attributable cost
Platform spend that can be traced to a team, workload or dataset rather than only to a bill.

Baseline
A dated finding set retained with its scope and window, against which the next assessment is compared.

Evidence Source Mapping

KOMTAS

Where each class of evidence is obtained, what it supports, and what the report says when it cannot be obtained. Filled in per client at scoping.

Query and workload history

Where it comes from
Warehouse query log / system views

What it is used for
Latency p95, concurrency, workload mix

If it is unavailable
Latency findings become interview-based; stated as such

Pipeline run history

Where it comes from
Orchestrator / scheduler history

What it is used for
Success rate, MTTR, load window use

If it is unavailable
Reliability is assessed from incident memory, not data

Billing and consumption records

Where it comes from
Platform billing export, workload tags

What it is used for
Cost per TB, idle compute, attribution

If it is unavailable
Cost findings stay at platform level, not workload level

Catalogue and lineage export

Where it comes from
Catalogue tool, or schema plus job parsing

What it is used for
Ownership, contract and lineage coverage

If it is unavailable
Coverage is sampled by hand on the critical list

Data model and DDL

Where it comes from
Schema export, repository, modelling tool

What it is used for
Modelling approach, historisation, duplication

If it is unavailable
Model is reconstructed from the physical schema only

Access control configuration

Where it comes from
Platform roles, groups and grants export

What it is used for
Access review coverage, masking, row-level rules

If it is unavailable
Security dimension is marked partial evidence

BI inventory and usage logs

Where it comes from
Reporting platform metadata and usage history

What it is used for
Certified vs. shadow, unused assets, trust signals

If it is unavailable
Consumption is assessed from interviews and sampling

Reconciliation output

Where it comes from
Existing controls, or a sample run during the engagement

What it is used for
Break rate, what is checked and what is not

If it is unavailable
A sample reconciliation is run in the window and labelled as such

Where it comes from

What it is used for

If it is unavailable

Access is requested once, early

Each of these is a separate approval in most organisations. Raising them in P0 as one list is the difference between a six-week and a twelve-week engagement.

Exports, not live access

Read-only exports over an agreed window are usually faster to approve than standing access and are sufficient for every measurement listed.

Unavailable is not unknown

Where a source cannot be obtained, the report names the source, the measurements it would have supported, and the resulting confidence level.

From Assessment to Delivery

KOMTAS

The assessment is the first stage of a sequence, not a standalone exercise. Each stage has an output the next one depends on.

STAGE 1 · Assessment

6 – 9 weeks

- Evidence collected, measured and placed
- Findings issued with evidence type and gap
- Target state decision made against criteria
- Dated baseline retained

RULE — Output is a decision, not a document

STAGE 2 · Foundations

1 – 2 quarters

- Ownership assigned on the critical list
- Lineage completed on the critical path
- Quality rules and reconciliation where they are missing
- Cost attribution made visible

RULE — Done before any platform change, whatever the target state

STAGE 3 · Platform work

2 – 4 quarters

- Refactor or migration executed in scoped waves
- Each wave carries its own acceptance criteria
- Old and new run in parallel only for a stated period
- Decommissioning dates set at the start

RULE — A parallel-run period with no end date becomes permanent double cost

STAGE 4 · Re-assessment

6 – 12 months after

- Same dimensions, same weights, same metric definitions
- Movement measured against the retained baseline
- Target levels revisited against current business need
- Remaining gaps re-sequenced

RULE — The method starts producing a trend only here

Stage 2 is the one most often skipped, because stage 3 is the one with a product attached to it. Ownership, lineage and quality gaps migrate with the data and arrive on the new platform with migration risk added to them.

Stage 4 is what makes stage 1 worth paying for twice. Without a re-assessment against the same definitions, there is no way to show that anything the programme did actually moved the platform.

The Limits of the Assessment and the Language Used

KOMTAS

Writing down what was not examined protects the value of what was. A reader who cannot see the boundary will assume there is none.

OUT OF SCOPE — STATED IN EVERY REPORT

Code and transformation logic review	Transformation correctness is not line-by-line verified; modelling approach is assessed, not every rule.
Data content and business correctness	Whether a figure is right in business terms is not in scope; whether it reconciles to source is.
Penetration testing	No active security testing is performed; configuration and review records are examined.
Vendor or licence negotiation	Contract terms and pricing positions are outside the assessment.
Full catalogue coverage	Lineage and ownership are assessed on the critical list, not across the whole estate.
Capacity sizing for a target platform	Sizing a replacement is a separate exercise with its own assumptions and inputs.

A good result in the assessed dimensions is not assurance for these areas.

THREE-STATE EVIDENCE SCALE

Evidence provided

A measurement, artefact or observation supports the stated level.

Partial evidence

Part of the evidence exists; the remainder was out of scope or could not be obtained.

Evidence not produced

The evidence could not be produced in the window. It does not mean the capability is absent.

Phrase not used: “the platform is non-compliant”. That is a judgement this engagement is not scoped to make.

HOW FINDINGS ARE WORDED

Language used

“The evidence for this control was not produced in this environment” rather than “this control does not exist”. The two are different claims and only one is supported.

Why it matters

An assessment is read by people who were not in the room. A sentence that overstates what was seen will be quoted back when it turns out to be wrong.

Confidence is stated

Where the measurement set was thin — short retention, no tagging, restricted access — the dimension carries a stated confidence level alongside its placement.

Comparability is the point

Same dimensions, same weights, same definitions, same evidence rules. Anything that changes between assessments is listed on the first page.

The assessment produces evidence for a decision. It does not certify the platform, it does not clear it against any standard, and it does not transfer responsibility for the decision away from the organisation that makes it.

SECTION 3

CLIENT FINDINGS

Everything up to this point is organisation-independent and is reused in every engagement. The pages in this section are filled in with the findings of one assessment and belong to one organisation only.

Current-state summary

Each dimension placed on the ladder, with evidence type, gap to target and the findings behind it.

Target state and rationale

The keep / refactor / replace decision against the criteria, with the rejected options and their reasons.

Delivery plan

Remediation themes in sequence with dependency, effort and risk, and what is deliberately not being addressed.

Client Findings — Current State by Dimension

KOMTAS

Each dimension with its placement, the evidence behind it and the gap to the target level. Filled in at the end of P2.

	DIMENSION	CURRENT	TARGET	EVIDENCE	KEY FINDING	CONFIDENCE
D1	Architecture and layering					
D2	Modelling and semantics					
D3	Data quality and reconciliation					
D4	Ingestion and pipeline reliability					
D5	Governance, lineage and metadata					
D6	Security, access and privacy					
D7	Performance and scalability					
D8	Cost and efficiency					
D9	Operations, DR and change					
D10	Consumption and BI enablement					
D11	AI and ML readiness					

Evidence column records the strongest evidence type behind the placement: measurement, artefact, observation or statement. A dimension placed on statements alone carries low confidence and is marked as such rather than left to read like the others.

Client Findings — Target State and Rationale

KOMTAS

The keep / refactor / replace decision, the criteria that drove it and the options rejected. Read together with the decision framework.

OPTION A

Keep and optimise

recommended / rejected — to be marked

OPTION B

Refactor in place

recommended / rejected — to be marked

OPTION C

Replace / lakehouse

recommended / rejected — to be marked

CRITERIA THAT DROVE THE DECISION

Which criteria pointed where, and which findings they rest on

REJECTED OPTIONS

Why each was rejected and under which condition it would be revisited

WHAT THE PLATFORM IS NOT THE CONSTRAINT FOR

Gaps that would follow the data onto any target platform

OPEN ITEMS AND ASSUMPTIONS

Unanswered scoping questions and the assumptions carried into the recommendation

The recommendation is not justified by the composite score. At least three criteria and the findings behind them are shown, and the two rejected options are given the same treatment as the recommended one.

Client Findings — Delivery Plan

KOMTAS

Remediation themes in sequence, with dependency, effort and risk, and what is deliberately not being addressed.

REMEDIATION THEME	DIMENSIONS ADDRESSED	START	END	DEPENDENCY AND RISK

EFFORT AND SEQUENCE RATIONALE	NOT BEING ADDRESSED	RE-ASSESSMENT POINT
Why this order; which dependencies forced it	What is deliberately out of the plan, and why	Agreed date, scope and the baseline it compares to

Every theme has an end date and a re-assessment point. Without them the plan has no way of showing that it worked, and the next assessment starts from opinion again rather than from a baseline.